

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CampusPulse – Student Engagement & Dropout Risk Analytics

Student Outcome Prediction and Explainability Using R

Case Study Report submitted in partial fulfillment of the Requirements for the Award of the Degree of
BACHELOR OF ENGINEERING
in
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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October 2026

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
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CERTIFICATE

This is to certify that the course case study report entitled “CampusPulse – Student Engagement & Dropout Risk Analytics” is being submitted by **Kaleru Sriniketh (160124771041)** in partial fulfillment for the award of the Degree of Bachelor of Engineering in Artificial Intelligence and Data Science to the Chaitanya Bharathi Institute of Technology, Hyderabad, carried out under my mentorship and supervision during the academic year 2026–27.

The results presented in this case study report have not been submitted to any other university or institute for the award of any degree.

Mrs. P. Greeshma
Assistant Professor

CampusPulse – Student Engagement & Dropout Risk Analytics

Student Outcome Prediction and Explainability Using R

1. Introduction and Problem Statement

Student continuation and academic success are important concerns for higher education institutions. Students may discontinue their studies or remain enrolled without completing their programmes within the observed period. Identifying patterns associated with these outcomes can help institutions understand student engagement and plan timely academic support.

CampusPulse is a data-driven case study developed in R using student academic, demographic, application and financial information. It analyses the factors associated with three observed outcomes—Dropout, Enrolled and Graduate—and compares classification models under two prediction scenarios. The project also uses feature importance and LIME to make model behaviour easier to interpret.

The project is intended as an analytical decision-support exercise. Its predictions should not be treated as definitive judgements about individual students or as a substitute for human review.

2. Objectives

• Explore the student dataset and describe the distribution of the three outcome categories.

• Prepare and clean the dataset for supervised classification.

• Compare Multinomial Logistic Regression and Random Forest models.

• Evaluate admission-time prediction (Scenario A) against prediction using first-semester information (Scenario B).

• Assess performance using accuracy, macro-precision, macro-recall, macro-F1 and confusion matrices.

• Interpret model behaviour using Random Forest feature importance and LIME.

3. Dataset

The study uses the UCI Machine Learning Repository dataset titled “Predict Students' Dropout and Academic Success” (Dataset ID 697). The dataset contains 4,424 student records and 37 columns, comprising 36 predictor variables and one target variable.

Attribute	Description
Dataset	Predict Students' Dropout and Academic Success
Source	UCI Machine Learning Repository, Dataset ID 697
Records	4,424 students
Columns	37 (36 predictors and 1 target)
Target	Dropout, Enrolled, Graduate
Missing values	0 in the dataset used
Dataset DOI	10.24432/C5MC89

The outcome distribution is shown below.

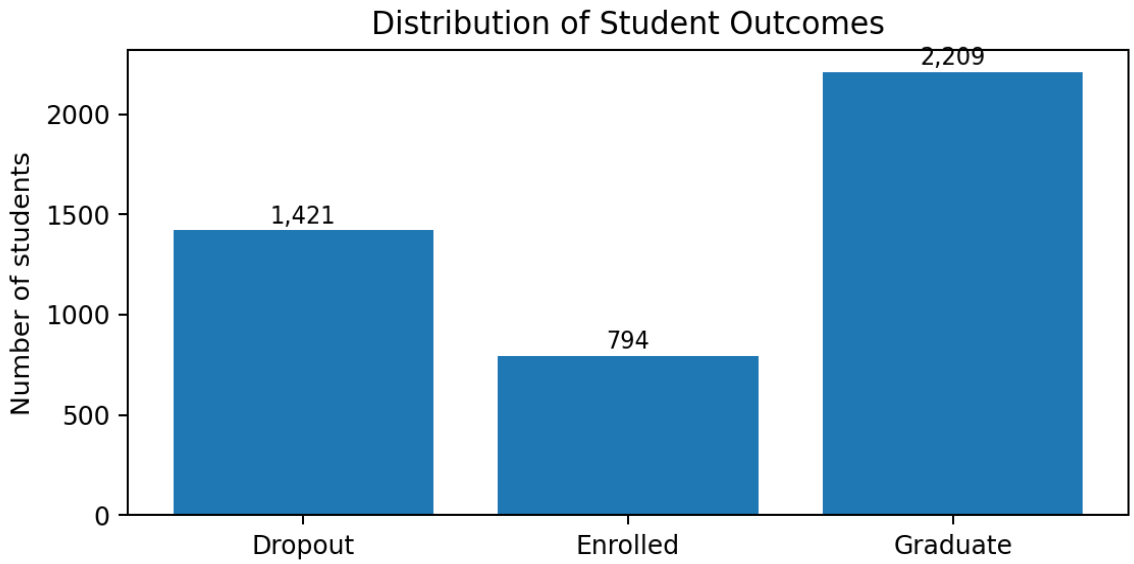


Figure 1. Distribution of student outcomes in the dataset.

4. Data Preparation

The original comma-separated dataset was imported into R. Column names were standardised using janitor::clean_names(). The dataset used in the analysis contained no missing values. Two prediction scenarios were created: Scenario A uses admission-time/background information, while Scenario B adds first-semester academic performance. Second-semester variables were excluded to avoid using later information in the prediction exercise.

Step	Description
Data import	Read UCI CSV dataset using comma separation.
Column names	Standardised using clean_names().
Missing-value check	No missing values found.
Target	Dropout, Enrolled and Graduate.
Scenario A	Admission-time/background predictors.
Scenario B	Scenario A plus first-semester academic performance.
Data split	Stratified train/test evaluation.

5. Exploratory and Statistical Analysis

5.1 Outcome distribution

Outcome	Students	Percentage
Dropout	1,421	32.12%
Enrolled	794	17.95%
Graduate	2,209	49.91%
Total	4,424	100.00%

5.2 Summary of selected variables

Outcome	Average age	Admission grade	First-semester grade
Dropout	26.1	125.0	7.26
Enrolled	22.4	126.0	11.1
Graduate	21.8	129.0	12.6

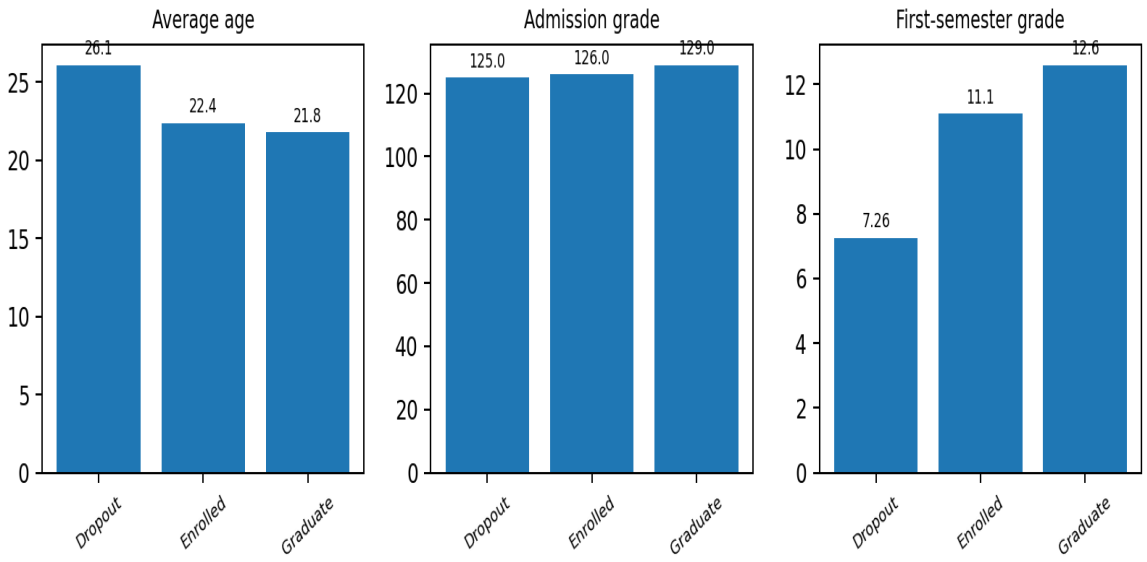


Figure 2. Average age, admission grade and first-semester grade by outcome.

The Dropout group has the highest average age at enrolment and the lowest average first-semester grade among the three groups. The Graduate group has the highest average admission grade and first-semester grade. These are descriptive associations and do not establish causal relationships.

6. Visualization

Visualizations were generated to communicate the outcome distribution, selected group averages, model performance and feature importance. Figures 1 and 2 present dataset-level patterns. Model comparison and feature-importance visualizations are presented in the Results section.

7. Prediction Models

7.1 Prediction scenarios

Scenario	Information available	Purpose
A	Admission-time/background information	Estimate outcome without first-semester performance.
B	Scenario A plus first-semester academic variables	Estimate outcome after early academic performance is available.

7.2 Algorithms

Multinomial Logistic Regression estimates probabilities across the three outcome categories. Random Forest combines multiple decision trees and can model non-linear relationships. Both algorithms were evaluated using a held-out test set and accuracy, macro-precision, macro-recall and macro-F1.

8. Results

8.1 Model performance comparison

Scenario	Model	Accuracy	Macro P	Macro R	Macro F1
A	Logistic Regression	64.21%	56.08%	53.59%	53.25%
A	Random Forest	65.80%	59.25%	54.24%	52.95%
B	Logistic Regression	73.61%	65.52%	63.92%	63.84%
B	Random Forest	74.97%	68.72%	64.13%	63.90%

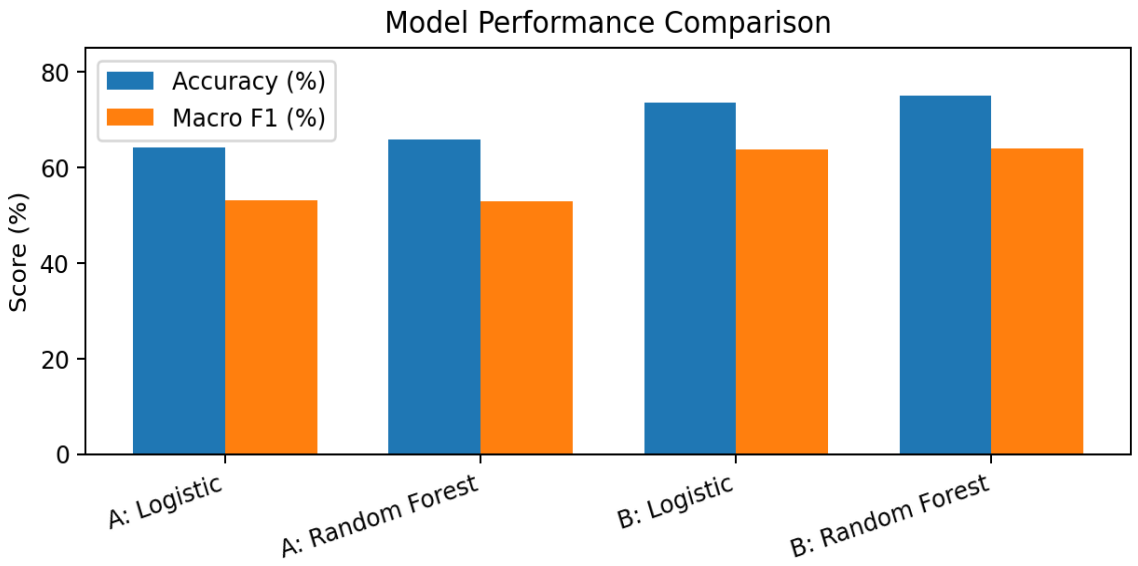


Figure 3. Accuracy and macro-F1 comparison across the four model configurations.

Scenario B Random Forest achieved the highest accuracy (74.97%) and macro-F1 (63.90%). Adding first-semester information improved accuracy by 9.17 percentage points for Random Forest and 9.40 percentage points for Logistic Regression. Macro-F1 shows that class-wise performance remains more challenging than overall accuracy alone suggests.

8.2 Class-wise evaluation

Rows represent predicted classes and columns represent actual reference classes.

Predicted / Actual	Dropout	Enrolled	Graduate
Dropout	227	52	27
Enrolled	16	34	13
Graduate	41	72	401

Class	Recall
Dropout	79.93%
Enrolled	21.52%
Graduate	90.93%

The model identifies Graduate and Dropout cases more effectively than Enrolled cases. Enrolled recall is 21.52%, indicating that many actual Enrolled records are assigned to another predicted category.

8.3 Feature importance

Rank	Feature	Importance
1	curricular_units_1st_sem_approved	337.54
2	curricular_units_1st_sem_grade	219.15
3	course	162.96
4	father_s_occupation	130.39
5	tuition_fees_up_to_date	111.06

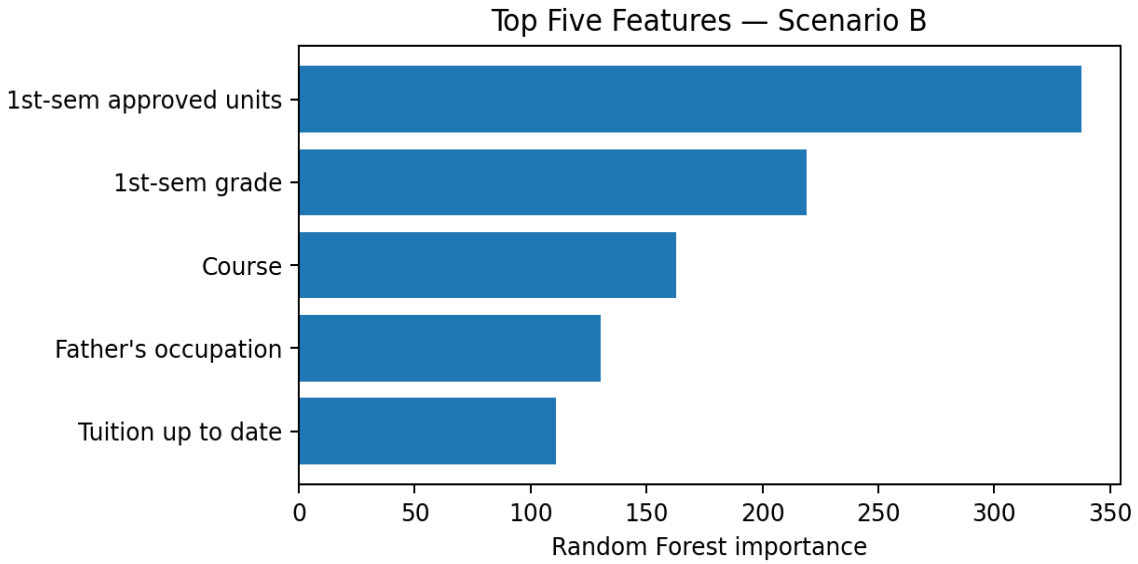


Figure 4. Top five Random Forest features in Scenario B.

First-semester approved curricular units and first-semester grade rank highest. Feature importance describes predictive contribution within this fitted model; it does not demonstrate causation.

8.4 LIME explanation

LIME was applied to one individual prediction from the Scenario B Random Forest. The model predicted Dropout with a model score of approximately 86.16%.

Feature	LIME weight	Direction
tuition_fees_up_to_date	0.2567	Supports Dropout
curricular_units_1st_sem_approved	0.2055	Supports Dropout
nacionality	-0.0975	Opposes Dropout
curricular_units_1st_sem_grade	0.1080	Supports Dropout
scholarship_holder	0.0662	Supports Dropout
debtor	-0.0490	Opposes Dropout
international	-0.0430	Opposes Dropout
educational_special_needs	-0.0285	Opposes Dropout

For this case, absence of approved first-semester curricular units and tuition-fee status were among the strongest contributions supporting the prediction. Binary values are encoded values; their natural-language meaning depends on dataset coding. The 86.16% score is a model output, not a guarantee.

9. Conclusion

CampusPulse demonstrates an end-to-end student outcome classification workflow using R, from data preparation and exploratory analysis to model comparison and explainability. Scenario B Random Forest achieved 74.97% accuracy and macro-F1 of 63.90%. Its class-wise recall was 79.93% for Dropout, 21.52% for Enrolled and 90.93% for Graduate. First-semester approved curricular units and first-semester grade were the leading Scenario B features. LIME provided a local explanation for one selected prediction.

Limitations

• Enrolled recall is substantially lower than recall for Dropout and Graduate.

• Results are based on one train/test evaluation and may vary with another split or external dataset.

• Feature importance and LIME describe model behaviour, not causal effects.

• The LIME explanation concerns one selected test record only.

• The model should support human review, not automate high-impact decisions about students.

Future Work

• Investigate class weighting and resampling to improve Enrolled-class recognition.

• Use repeated cross-validation and, where possible, an external institutional dataset.

• Test probability calibration and threshold-based approaches for early support.

• Validate explanations with academic advisors and domain experts.

10. References

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